

ROHIT PENTI

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA
Master of Information Systems Management

August 2024 – August 2025

- Relevant Courses: Cloud Computing, Object Oriented Programming in Java

Indian Institute of Technology, Kanpur, India

July 2015 – April 2020

Bachelor & Master of Science (Integrated) in Mathematics & Scientific Computing

- Relevant Courses: Data Structure & Algorithms, Parallel Algorithms, Theory of Computation, Probability & Statistics, Statistical Inference, Time Series Analysis, Markov Chain Monte Carlo

SKILLS

Languages: C, C++(Advanced), Python, Java, R, MATLAB, SQL, Bash

Tools & Tech: AWS, Azure, GCP, Apache Maven, Git, SQLite, Splunk, Terraform, Kubernetes, Docker, Spark, Jupyter, Zeppelin, JUnit, GMock

Knowledge: Multithreading & Concurrency, Relational Databases, Microservices

EXPERIENCE

ADOBE SYSTEMS, Bangalore, India

Software Development Engineer II

February 2022 – July 2024

- Owned and led the development of new features in CoreSync, a file synchronization process, integrating with Adobe's backend APIs
- Designed and implemented garbage collection for CoreSync's local SQLite database, ensuring data integrity during cleanup operations and reclaiming approximately 500GB of storage across client machines. Demonstrated expertise in SQLite by designing efficient schemas
- Engineered a notification-based synchronization mechanism, replacing a polling-based model, reducing cloud services' load by over 75%
- Designed and maintained Splunk dashboards to monitor feature performance, enabling faster bug resolution and decision-making process
- Adopted feature flag service, enabling phased rollouts and selective feature toggling for user groups, streamlining production rollouts
- Mentored junior developers in advanced C++ programming, design principles, and debugging techniques to ensure high-quality code

Software Development Engineer

August 2020 – Jan 2022

- Refined sync logic for complex conflict scenarios, leveraging advanced visitor patterns to boost successful sync rates from 97% to 98.3%
- Enhanced the resiliency of sync engine by debugging and resolving crashes related to thread concurrency, significantly improving stability
- Authored and maintained comprehensive functional and unit tests using the GMock framework to ensure correctness of the sync engine

ACCENTURE, Bangalore, India

Applied Intelligence Software Intern

May 2019 – July 2019

- Enhanced Google Search-based web crawler by integrating Wikipedia and Google Maps, improving SMB website accuracy from 25% to 80%
- Built a generic contact crawler to extract demographic data and proposed a solution for a real-time search tool tailored for SMB discovery

FIBE (formerly EARLYSALARY), Pune, India

Data Science Intern

May 2018 – July 2018

- Created a logistic regression-based application scorecard to predict delinquency rate. Extracted and cleaned over 1,000 features from SQL, applying feature selection techniques like recursive feature elimination and bucketizing variables. Achieved 92% accuracy in the model

ACADEMIC PROJECTS

Cloud Microservices Engineering

- Developed and optimized three microservices (Blockchain, QR Code, and Twitter Service) from scratch using Kubernetes, Helm, and Terraform, achieving target throughput and cost efficiency
- Implemented core functionalities in Java, writing logic for the application and integrating Amazon RDS for efficient data management in the Twitter Service

AWS Autoscaling and Elasticity

- Implemented autoscaling on AWS using Terraform, configuring ELB and ASG to handle dynamic loads, ensuring high availability and cost-efficiency. Monitored CloudWatch metrics to manage scaling and maintain Quality of Service
- Analyzed patterns and optimized policies to balance performance and budget constraints, setting up health checks and for fault tolerance

Kubernetes and Microservices

- Deployed a scalable chat application using Docker and Kubernetes, managing clusters on GCP and Azure with Helm charts
- Automated CI/CD with GitHub Actions, implementing autoscaling and multi-cloud deployments for high availability and fault tolerance

Batch Processing with Spark and Scala

- Performed ETL on a 10.4GB Twitter social graph using Apache Spark, implementing the PageRank algorithm to identify influential users
- Extended PageRank to develop user recommendation system, leveraging Spark's RDD and DataFrame APIs for efficient data processing

Cloud Storage and Databases

- Evaluated SQL and NoSQL databases for a social network application, implementing MySQL indexing and MongoDB queries for optimization
- Deployed a scalable database architecture using Azure CosmosDB, MySQL, and MongoDB, integrating them with Java APIs for data retrieval

Intelligent Taxi Fare Predictor

- Designed a scalable application integrating Vertex AI, Vision API, and Speech APIs to predict taxi fares from voice commands and landmark images, deploying on Google App Engine with hybrid landmark recognition (AutoML image classification model) and geocoding capabilities

Natural Language Query to SQL Query Translation

- Built and trained a SEQ2SQL model using a 2-layered LSTM and Reinforcement Learning, achieving 55.4% accuracy on the WikiSQL dataset
- Demonstrated advanced deep learning capabilities by translating natural language queries into SQL queries